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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: CLOUD BASED DATA ANALYTICS SPECIALITY (23SDCS08A) **A.Y. 2025 – 2026**

Section – 4

Title of The Project	CLOUD BASED CUSTOMER BEHAVIOR ANALYTICS SYSTEM	
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The Cloud-Based Customer Behavior Analytics System is designed to analyze, process, and visualize customer transaction data using a streamlined and cost-effective cloud architecture. The primary objective of this project is to extract meaningful business insights—such as purchasing patterns, customer segmentation, and spending behavior—by applying data analytics techniques to real-world transactional data.

The system begins with data processing in Google Colab, where raw customer transaction data is imported and undergoes several preprocessing steps. These include data cleaning (handling missing or inconsistent values), transformation (formatting and structuring data for analysis), and analytical computations (such as aggregations, trend analysis, and feature generation). This stage ensures that the dataset is accurate, structured, and ready for further analysis.

Once processed, the refined dataset is stored in Azure Blob Storage, which serves as a scalable, secure, and cost-efficient cloud-based storage solution. This allows the system to handle large volumes of data while maintaining accessibility and flexibility for future use or integration with other tools.

For visualization and reporting, Power BI is used to build interactive dashboards. These dashboards provide a clear and dynamic representation of customer behavior, including key metrics such as sales trends, customer lifetime value, segmentation groups, and spending distributions. Users can interact with the dashboards to filter data, explore patterns, and gain actionable insights that support strategic decision-making.

The system follows a structured pipeline approach: analytics first, cloud integration second, and business insights last. This design prioritizes efficient data processing before leveraging cloud storage and visualization tools. By avoiding heavy cloud computing services and utilizing only essential Azure resources, the solution remains budget-friendly while still being scalable and industry-relevant.

Overall, the project demonstrates how organizations can implement a practical and efficient customer analytics system using minimal resources. It highlights the importance of data-driven decision-making and provides a foundation for businesses to better understand customer behavior, optimize marketing strategies, and improve overall performance.